



TASHKENT UNIVERSITY OF TECHNOLOGY

Faculty of Energy Engineering

1. Subject/Module Information

Field	Content
Subject/Module Name	Mathematical Analysis – I
Subject/Module Code	CC-MAAN10101
Subject/Module ECTS	6
Academic year	2026 - 2027
Study Semester	Fall (semester 1)
Subject/module type	Mandatory
Program Affiliation	Bachelor of Science for all tracks
Module Coordinator(s)	Dr. Georgios Neofotistos Mr. Otabek Mirzaxmedov

2. Forms of Learning and Teaching

Workload breakdown (30 hrs/ECTS convention):

Component	Hours
Total Classroom Hours	90
Lecture	60
Practical Classes (Laboratory, Workshop)	30
Independent Study	90
Total Workload	180

3. Module Description

The aim of the course is to provide engineering students with a solid and coherent foundation in Mathematical Analysis, focusing on the tools and reasoning most essential for scientific and engineering applications. Students will learn how to use mathematical language and symbolic computation with awareness and precision. The course covers number systems, functions (of one variable), limits, continuity, derivatives, integrals, and their applications. The course emphasizes conceptual understanding, mathematical reasoning, analytical problem solving, computational thinking, and engineering applications.

4. Entry Requirements

Field	Content
Prerequisites	No
Co-requisites	No
Additional Remarks	No

5. Intended Learning Outcomes (ILOs)

Each ILO carries a Bloom level and is mapped to ASIIN's four competence dimensions — Subject-specific (Fachkompetenz), Methodological (Methodenkompetenz), Systemic (Systemkompetenz), Social (Sozialkompetenz).

No	Bloom Level	ILO	ASIIN Dimension
1	Understand	Explain the key properties of real and complex numbers, key concepts of functions (including elementary functions, transformations, compositions, and inverse functions), limits,	Subject-specific

		asymptotics, continuity, derivatives and differentiability, integrals, integrability, and Riemann integration.	
2	Analyze	Analyze functions (of one real variable) by examining their limits, continuity, discontinuities, asymptotic behavior, derivatives, extrema, monotonicity, convexity, inflection points, integrals and integrability, and interpret their properties algebraically and geometrically.	Subject-specific
3	Select	Select appropriate analytical methods to calculate limits, derivatives, indeterminate forms, extrema, and integrals (e.g. change of variables or integration by parts) to solve given mathematical or engineering problems.	Methodological
4	Evaluate	Evaluate mathematical reasoning and solutions by checking whether the conditions of the relevant analytical methods or theorems are satisfied, assessing approximation errors, and determining whether the selected methods and results are appropriate within each problem's context.	Systemic
5	Apply	Collaborate in small groups on assigned, engineering-oriented, calculus projects, and clearly communicate mathematical reasoning, solutions and results, in written and verbal form, to peers.	Social

6. Course Content (Weekly Breakdown)

Week	Topic	Format	Hours
1	Introduction to the course. Number systems: natural, integer, rational, real, and complex numbers.	Lecture + Practical	6
2	Functions of one real variable. Domain, range, elementary functions, transformations, composition and inverse functions. Graphing functions.	Lecture + Practical	6
3	Limits of functions. Intuitive and ε - δ definitions of limits. One-sided and infinite limits.	Lecture + Practical	6
4	General properties of limits. Standard limits. Indeterminate forms. Asymptotic equivalence and Landau symbols. Test 1.	Lecture + Practical + Assessment	6
5	Continuous functions. Continuity at a point and at an interval. Discontinuity points. Zeros and intermediate value theorems. Weierstrass' Theorem. Algebra of continuous functions. Continuity of compositions and inverse functions.	Lecture + Practical	6
6	Derivatives and differentiability: Definition and geometric interpretation of the derivative. Differentiability and continuity. Left- and right-hand derivatives. Kink points, cusps, and vertical tangents. Higher derivatives. Differentiation rules (product, quotient)	Lecture + Practical	6
7	Derivatives of elementary functions: derivatives of polynomial, rational, exponential, logarithmic, trigonometric functions.	Lecture + Practical	6
8	Midterm Assessment	Assessment	6
9	Differentiation rules (chain rule). Algebra of derivatives. Derivatives of composite and inverse functions. Implicit differentiation.	Lecture + Practical	6
10	Differentials, linear approximation, and applications. Local and global maxima and minima. Critical and stationary points. Fermat's theorem. Rolle's theorem. Lagrange's and Cauchy's Theorems.	Lecture + Practical	6
11	First-derivative tests. de l'Hospital's theorem and evaluation of indeterminate forms. Convexity and concavity.	Lecture + Practical	6
12	Second-derivative tests, inflection points, asymptotes, and systematic curve sketching. Taylor's formula. Remainder and approximation error.	Lecture + Practical	6
13	Integrals. The Riemann integral and its basic properties. Riemann sums and integration. The Dirichlet function. Integrability of continuous functions. Test 2	Lecture + Practical + Assessment	6
14	Fundamental Theorem of the integral calculus. Antiderivatives and evaluation of definite integrals. Change of variables and integration by parts.	Lecture + Practical	6
15	Applications of integrals to engineering problems. Course review.	Lecture + Practical	6

7. Independent Study Assignments*

Week	Assignments	Hours
1	Individual problem set on how natural, integer, rational, real, and complex numbers relate; algebraic operations with complex numbers (conversion among algebraic, trigonometric, and exponential forms); applying the de Moivre's formula and calculating complex roots.	6
2	Individual problem set on computing and visualizing elementary functions and inverse functions (graphical visualization of functions using Python or other mathematical software).	6
3	Individual problem set on the intuitive and formal definition of limits, infinite limits, limits at infinity, and their graphical interpretation.	6
4	Individual problem set on standard limits, indeterminate forms, asymptotics and Landau symbols. Review and self-assessment exercises in preparation for Test 1.	6
5	Individual problem set on continuity at a point and on an interval, classification of discontinuities, zeros of continuous functions, the Intermediate Value and Weierstrass theorems, and continuity of compositions and inverse functions.	6
6	Individual problem set on the definition and geometric interpretation of the derivative, and on identification of kink points, cusps, and vertical tangents.	6
7	Individual problem set on derivatives of selected polynomial, rational, exponential, logarithmic, and trigonometric functions, and the interpretation of the derivatives as rate of change in engineering applications.	6
8	Individual self-assessment: systematic review of number systems, functions, limits, continuity, and introductory differentiation; in preparation for the Midterm Assessment.	6
9	Individual problem set on differentiation rules (product, quotient, and chain rules); on derivatives of composite and inverse functions; implicit differentiation and higher-order derivatives.	6
10	Individual problem set on local and global extrema, critical and stationary points, Fermat's theorem, Rolle's theorem, and the Lagrange and Cauchy Mean Value Theorems.	6
11	Individual problem set on applying de l'Hospital's theorem on indeterminate forms.	6
12	Individual problem set on second-derivative tests to identify inflection points and asymptotes; and on applying the Taylor formula on selected functions, identifying the remainder terms, and calculating the approximation errors.	6
13	Individual problem set on the Riemann integral, practicing on partitions, and upper and lower sums; explaining Riemann integrability and the Dirichlet function. Review and practice for Test 2.	6
14	Individual problem set on the Fundamental Theorem of Calculus and antiderivatives, evaluating definite integrals, by change of variables and by integration by parts.	6
15	Individual problem set on modeling an engineering system using differential and integral calculus. Wrap-up review for the Final Exam.	6

***Note:** Independent study assignments for the course (module) should incorporate a variety of learning tasks. At least one-third (1/3) of these assignments should require collaborative learning in small groups.

8. Teaching and Learning Methods

- Lectures
- Seminars / laboratory classes
- Homework and group projects
- Discussions
- Testing (written or computer based)

9. Assessment Methods and Completion

Component	Examination Type	Duration/Length	Weight (%)	ILOs
Midterm Exam	Paper-based problem solving	1–2 hours	30	1, 3
Test 1	Paper-based problem solving	1–2 hours	15	1, 3
Test 2	Paper-based problem solving	1–2 hours	15	2
Final Exam	Paper-based problem solving	1–2 hours	40	1–4

Grading and Completion

89-100 points = A - Excellent
75-89 Points = B - Good
60-74 Points = C - Satisfactory
0 - 59 Points = D - Poor

10. Indicative Literature

- Strang, G., & Herman, E. (2016). *Calculus Volume 1*. OpenStax, Rice University. This textbook is accessible for free at <https://openstax.org/details/books/calculus-volume-1>
- Strang, G. (1991). *Calculus* (updated 3rd ed.). Wellesley-Cambridge Press. This textbook is accessible for free at https://ocw.mit.edu/courses/res-18-001-calculus-fall-2023/resources/mitres_18_001_f17_full_book_pdf/ (through MIT OpenCourseWare)
- Prof. Neofotistos' Course Notes (weekly hand-outs, uploaded on Moodle)

11. Statement on Plagiarism and Academic Dishonesty:

Tashkent University of Technology observes a zero tolerance policy regarding academic dishonesty. Per university policy as described in the academic catalog, all cases of academic dishonesty will be reported. The following policy will be used for this course:

1. First instances of academic dishonesty will result in a 0 grade on the assignment, but a second attempt may be allowed at the instructor's discretion.
2. Second instances of academic dishonesty will result in a 0 grade on the assignment with no opportunity to reattempt.
3. Third instances of academic dishonesty will result in automatic failure of the course.

12. Statement on the use of AI-tools

Tashkent Tech recognizes artificial intelligence (AI) as a powerful tool for learning, innovation, and skill development. Students are encouraged to use generative AI tools (such as ChatGPT, Claude, DeepSeek or similar platforms) to clarify concepts, debug code, and improve their learning through dialogue (with the AI tools). When used responsibly, AI can serve as a valuable collaborative partner. However, any content generated by AI must be treated as a source, meaning that students must explicitly cite the tool used and clearly document how it assisted you in their workflow, in accordance with each course's specific guidelines.

Academic integrity remains the fundamental principle governing AI use. Students may not submit AI-generated text, code, calculations, or other outputs as their own work unless explicitly authorized by the instructor. All graded assignments, essays, and exams must reflect the student's own understanding and be explainable in detail upon request. The unauthorized or unacknowledged use of AI to complete assignments will result in academic penalties, as stated in the *Statement on Plagiarism and Academic Dishonesty* guidelines.

Attendance Requirements: Per University policy, students must attend at least 75% of class meetings to be allowed to take the final exam. If you miss 25% or more of class meetings, you will not be allowed to take the final exam.